LHCb Open Data Analysis of Three-Body B-Meson Decays

Invariant-mass reconstruction, signal classification, CP-asymmetry diagnostics and Dalitz-plot studies

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HEP portfolio project

October 26, 2026

Physics Motivation

- The visible Universe is matter dominated.
- CP violation is required but not yet sufficient in the Standard Model.
- Heavy-flavour decays probe weak phases and hadronic interference.

Why B mesons?

- many decay channels
- displaced vertices
- rich multibody phase space

LHCb Experiment and Dataset

- LHCb: forward spectrometer at the LHC.
- Optimized for beauty and charm hadrons.
- Excellent tracking, vertexing and particle identification.

Input sample

- CERN Open Data record 4900 (LHCb, 2017, CC0)
- ROOT files, `DecayTree` TTree
- track momenta, charge, PID, vertex variables
- 5.1×10^6 candidates

Analysis Workflow

ROOT → `uproot` → `pandas` → reconstruction → cuts → ML → physics observables

Data

Branch loading
candidate tables
reproducible paths

Physics

invariant mass
event selection
charge split

Outputs

figures
ML diagnostics
CP and Dalitz studies

Invariant-Mass Reconstruction

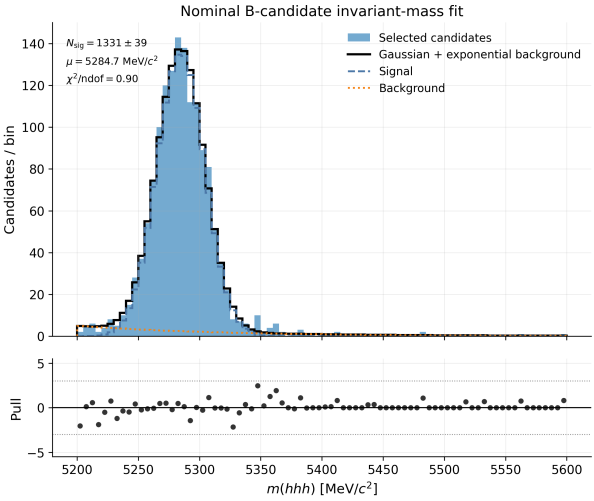
Three-kaon hypothesis

$$E_i = \sqrt{p_{x,i}^2 + p_{y,i}^2 + p_{z,i}^2 + m_K^2}$$

$$m_B^2 = \left(\sum_i E_i \right)^2 - \left| \sum_i \vec{p}_i \right|^2$$

- assign $m_K = 493.677\ \mathrm{MeV}/c^2$
- build daughter four-vectors
- sum to reconstruct B candidate
- vectorized in `numpy`/`pandas`

Mass Spectrum and Signal Fit



$$f(m) = N_s\, G(m;\mu,\sigma) + N_b\, E(m;\alpha)$$

- extended binned likelihood, MINUIT
- $N_s = 1331 \pm 39$
- $\mu = 5284.7 \pm 0.6$ MeV/c^2
- $\sigma = 19.5 \pm 0.5$ MeV/c^2
- $\chi^2/\mathrm{ndof} = 0.90$

Fitted from 5200 MeV/c^2 up: below that lies a partially reconstructed structure the background model does not describe.

Event Selection

Baseline cuts

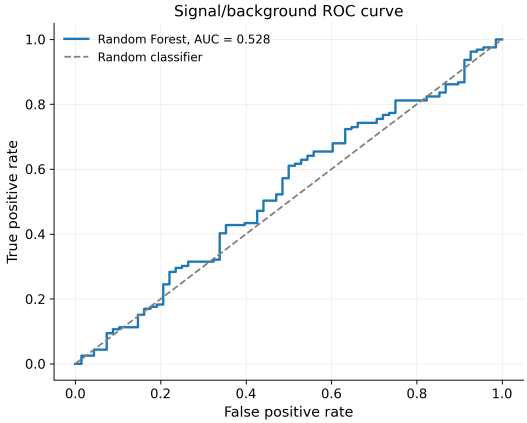
$$\chi^2_{\text{vtx}}(B) < 5$$

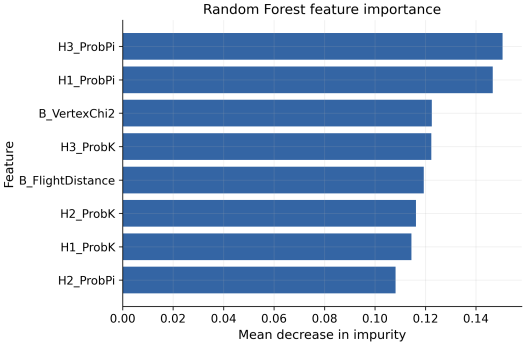
$$d_{\text{flight}}(B) > 20$$

$$P_K(H_1), P_K(H_2), P_K(H_3) > 0.5$$

- vertex quality: common displaced decay point
- flight distance: B lifetime signature
- kaon PID: consistency with $B \to KKK$
- same cuts reused across notebooks

Signal/Background Classification





- Sideband-labelled Random Forest; mass excluded from the features.
- $\mathrm{AUC} = 0.528$: **no separation**. A negative result, reported as one.

The baseline cuts already removed the background the available features could discriminate, and only 755 labelled candidates remain. The classifier is not used in any result that follows.

Raw Charge Asymmetry

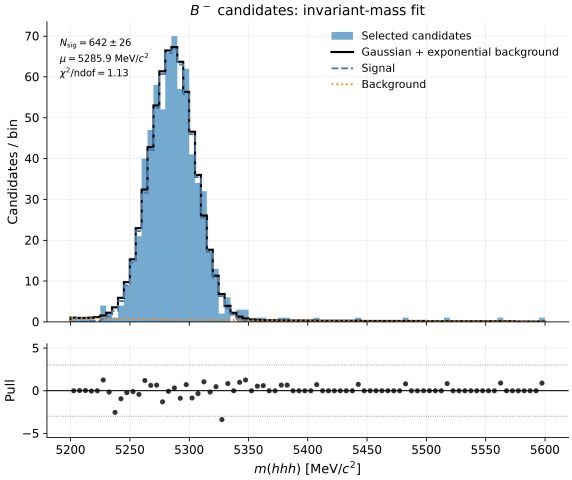
$$A_{\text{raw}} = \frac{N(B^-) - N(B^+)}{N(B^-) + N(B^+)}$$

- yields from *separate fits*, not window counting
- $N(B^+) = 689 \pm 29$, toy fit p-value 0.035
- $N(B^-) = 642 \pm 26$, toy fit p-value 0.43

Result

$$A_{\text{raw}} = -0.035 \pm 0.029\,(\text{stat.}) \pm 0.044\,(\text{syst.})$$

Raw: no detector, production, trigger or efficiency corrections; single magnet polarity.



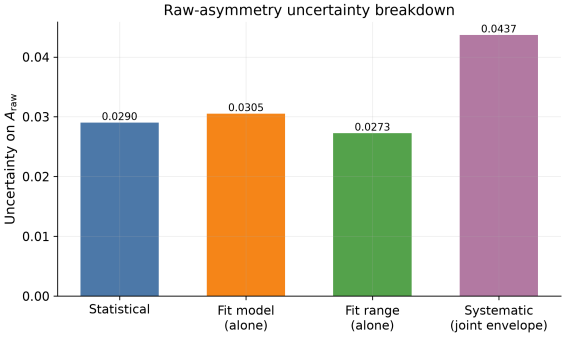
Validation and Systematics

2000 toy experiments

- generated from the fitted model, refitted
- pull width 0.9999, bias $< 10^{-4}$
- no fit failures
- quoted statistical error is calibrated

Systematics

- model alone 0.031, range alone 0.027 (Barlow)
- not independent: one low-mass shoulder in B^+
- joint model $\times$ range envelope: 0.044

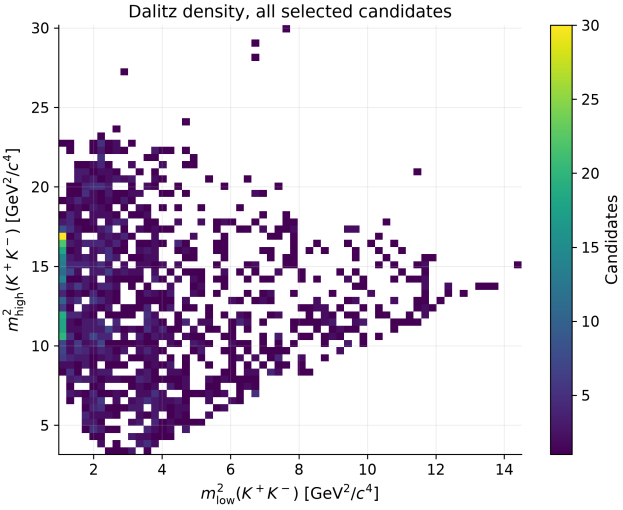


Configurations accepted only if both charge fits reach a valid minimum with $\chi^2/\mathrm{ndof} < 2.5$ — criteria fixed in advance. All 20 model $\times$ range configurations pass; the range sensitivity exists only for the nominal model.

Dalitz-Plot Analysis

- Pair masses
  $m_{ij}^2 = (E_i + E_j)^2 - |\vec{p}_i + \vec{p}_j|^2$
- Track labels carry no meaning: pairs found from the charges
- Opposite-sign pairs, folded:
  $m_{\text{low}}^2(K^+K^-) \leq m_{\text{high}}^2(K^+K^-)$

Same coordinates for B^+ and B^- under CP; the $\phi(1020)$ band sits at $m_{\text{low}}^2 \approx 1.04\ \text{GeV}^2$.



Limitations and Future Work

Current limitations

- raw asymmetry, no corrections
- single magnet polarity
- low-mass structure excluded, not modelled
- classifier does not separate

Next steps

- model the partially reconstructed component
- combine MagnetUp/MagnetDown
- control channel for production/detection asymmetries
- background subtraction per Dalitz bin

Conclusion

- Reproducible HEP workflow from ROOT ntuples to physics plots.
- Invariant-mass reconstruction under a kaon mass hypothesis.
- Shared selections and modular Python functions.
- Likelihood fits, toy validation, systematics and Dalitz diagnostics.
- $A_{\mathrm{raw}} = -0.035 \pm 0.029\,(\mathrm{stat.}) \pm 0.044\,(\mathrm{syst.})$, raw and uncorrected.

A compact portfolio analysis mirroring a real LHCb workflow